University of Dhaka

Assignment - 2

Course Title: Advanced Econometrics-II
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Question:

A data analyst estimated an ARMA (2,1) model for changes in the clothing industry's profits. However, the true model is actually an ARMA (2,2). Use simulations to show that the data analyst will estimate a biased model.

Solution:

Let's assume an ARMA (2,2) model is,

$$y_t = 0.4y_{t-1} + 0.3y_{t-2} + \varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-2}$$

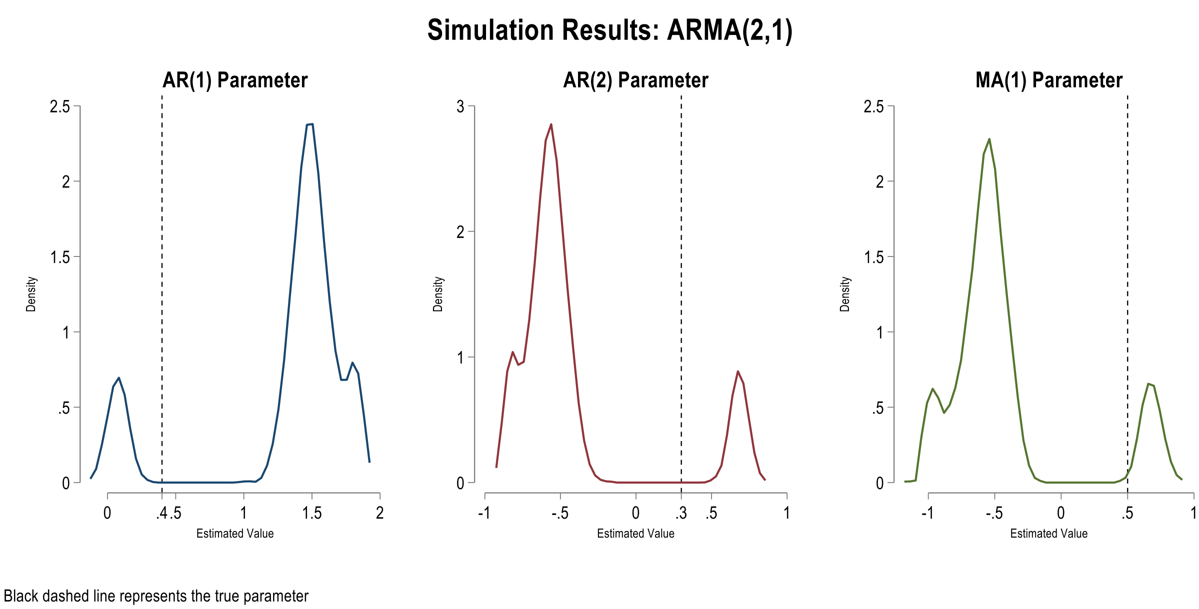
Where the coefficient of AR (1) = 0.4

coefficient of AR (2) = 0.3

coefficient of MA (1) = 0.5

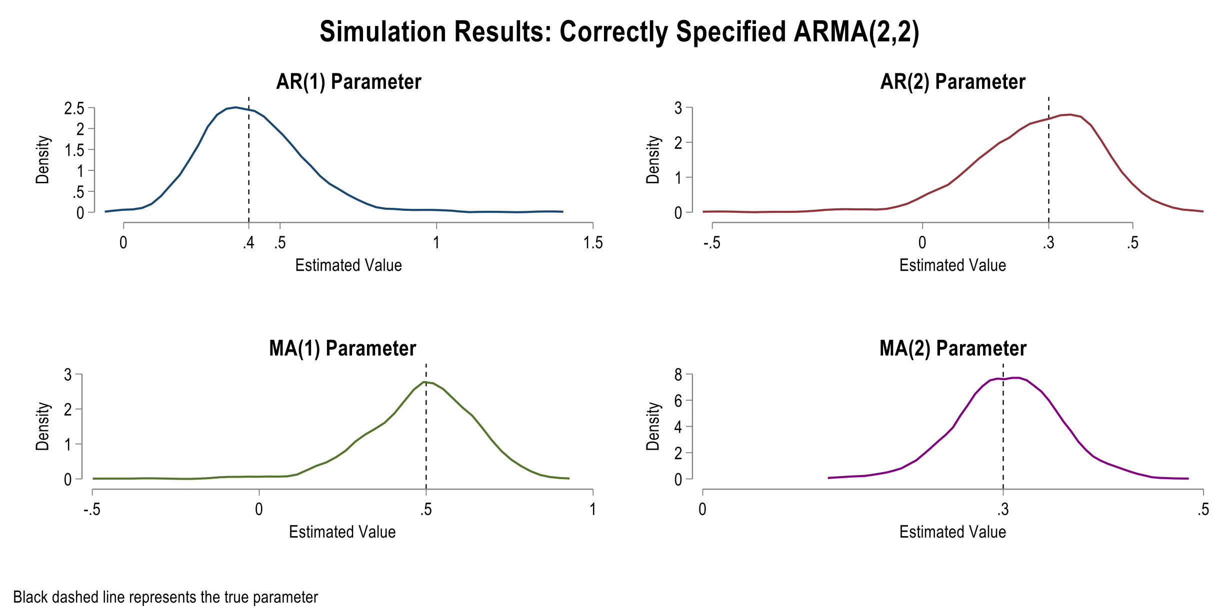
coefficient of MA (2) = 0.3

Now if a data analyst estimated an ARMA (2,1) model then the estimated coefficients will be biased downward or upward. To show this we will run a simulation with 1000 samples. With each sample we will estimate the coefficient, and then we will compare the expected value or mean of coefficients with the true parameters.



Coefficients	True value	Estimated Value	Biased
AR (1)	0.4	1.32	Upward by 0.92 points
AR (2)	0.3	-0.42	Downward by 0.72 points
MA (1)	0.5	-0.43	Downward by 0.93 points

From the diagram we can see that the estimated values exhibit multimodal distribution and does not converge to the single true parameters. The AR (1) parameter is estimated biased upward and both the AR (2) and MA (1) parameters are estimated biased downward.



Now if the data analyst had correctly specified the ARMA (2,2) model then the simulation with 1000 samples would yield similar estimated values close to true parameters and their distribution would follow the same true parameters which we can see from diagram for correctly specified ARMA (2,2) model.

In conclusion, a miss specified ARMA (2,1) model would yield biased coefficients for each value when the true model is ARMA (2,2).